Conversion from one Base to Another

1. Convert the decimal number 156 to:

- **Binary**

$$156_{10} = 10011100_2$$

- **Octal**

$$156_{10} = 234_8$$

- **Hexadecimal**

$$156_{10} = 9C_{16}$$

2. Convert the binary number 101101 to:

- **Decimal**

$$1 * 2^5 + 0 * 2^4 + 1 * 2^3 + 1 * 2^2 + 0 * 2^1 + 1 * 2^0$$
$$32 + 0 + 8 + 4 + 0 + 1$$
=45

- **Octal**

Convert each group:

$101_2 = 5_8$

$101_2 = 5_8$

= 55_8

- **Hexadecimal**

Convert each group:

$0010_2 = 2_{16}$

$1101_2 = D_{16}$

= $2D_{16}$

3. Convert the octal number 745_8:

- **Decimal**

$$7 * 8^2 + 4 * 8^1 + 5 * 8^0$$
$$7 * 64 + 4 * 8 + 5 = 448 + 32 + 5$$
$$= 485$$

- **Binary**

  Convert each octal digit to 3-bit binary:

  7 → 111

  4 → 100

  5 → 101

  Combine:

  $745_8 = 111\ 100\ 101_2$

  $= 111100101_2$

- **Hexadecimal**

  Use the binary result:

  Binary: 111100101

  Pad to groups of 4:

$$0011\ 1100\ 0101$$

  Convert each group:

  0011 → 3

  1100 → C

  0101 → 5

  = $3C5_{16}$

4. Convert the hexadecimal number $3F9_{16}$:

- Decimal

$$3 * 16^2 + F * 16^1 + 9 * 16^0$$

Remember: F = 15

$$3 * 256 + 15 * 16 + 9 = 768 + 240 + 9$$
$$= 1017$$

- Binary

Convert each hex digit to 4-bit binary:

3 → 0011

F → 1111

9 → 1001

Combine:

$$3F9_{16} = 0011\ 1111\ 1001_2$$
$= 001111111001_2$

- Octal

Use the binary result:

Binary:

$$001111111001$$

Group into 3 bits:

$$001\ 111\ 111\ 001$$

Convert each:

001 → 1

111 → 7

111 → 7

001 → 1

$= 1771_8$